

Nonlinear Registration

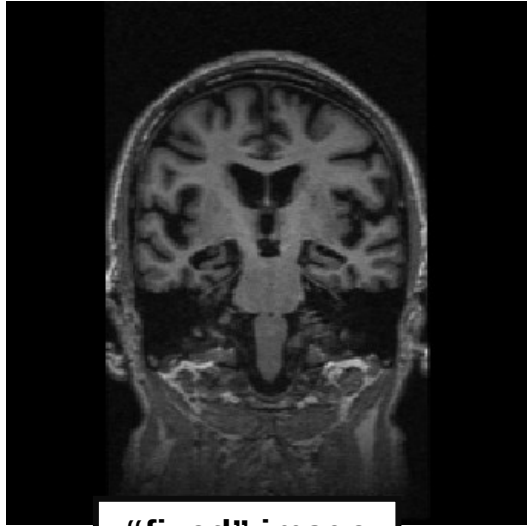


Aalto-yliopisto
Aalto-universitetet
Aalto University

Medical Image Analysis

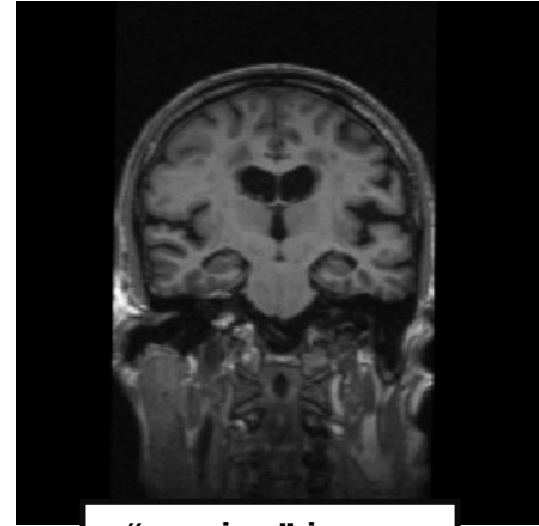
Koen Van Leemput

Spatial transformations



“fixed” image

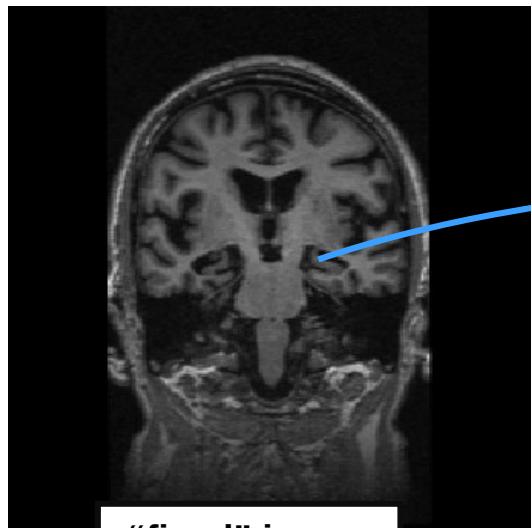
$$\mathbf{x} = (x_1, \dots, x_D)^T$$



“moving” image

$$\mathbf{y} = (y_1, \dots, y_D)^T$$

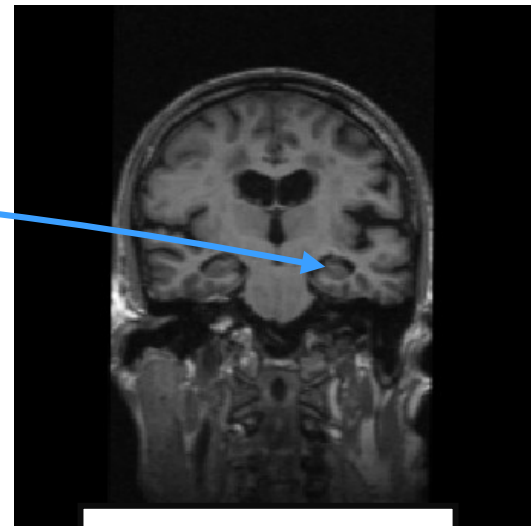
Spatial transformations



"fixed" image

$$\mathbf{x} = (x_1, \dots, x_D)^T$$

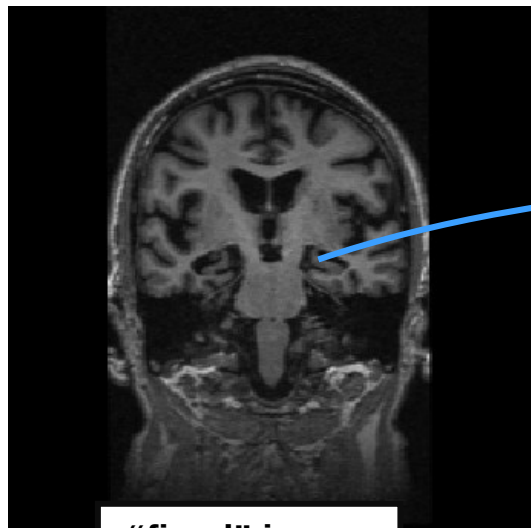
$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \begin{pmatrix} y_1(\mathbf{x}, \mathbf{w}) \\ \vdots \\ y_D(\mathbf{x}, \mathbf{w}) \end{pmatrix}$$



"moving" image

$$\mathbf{y} = (y_1, \dots, y_D)^T$$

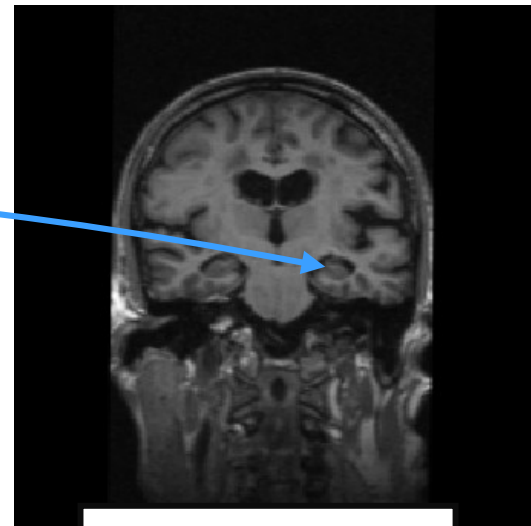
Spatial transformations



"fixed" image

$$\mathbf{x} = (x_1, \dots, x_D)^T$$

$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \begin{pmatrix} y_1(\mathbf{x}, \mathbf{w}) \\ \vdots \\ y_D(\mathbf{x}, \mathbf{w}) \end{pmatrix}$$



"moving" image

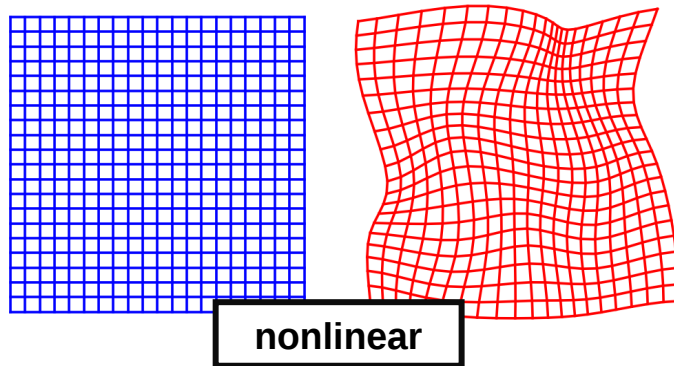
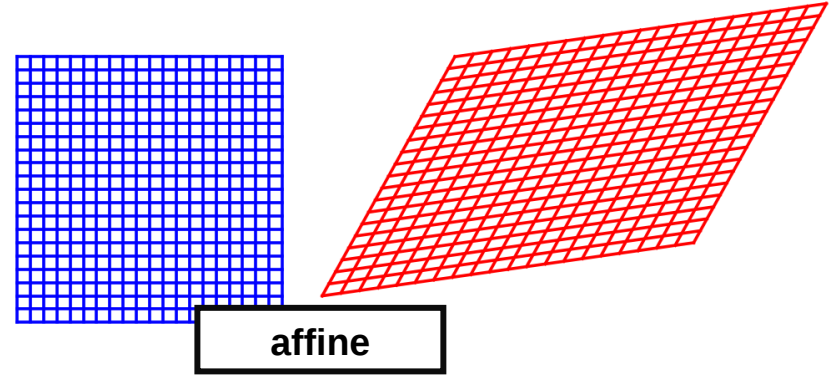
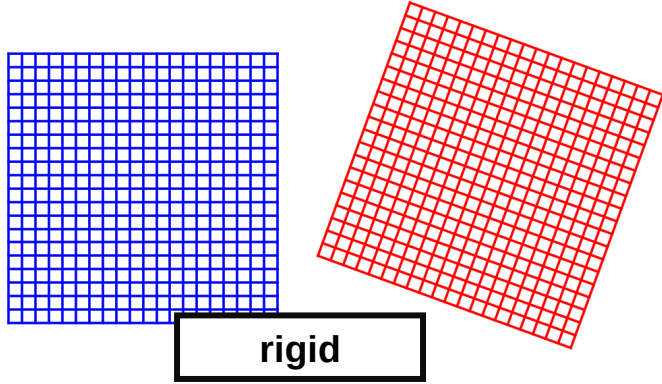
$$\mathbf{y} = (y_1, \dots, y_D)^T$$



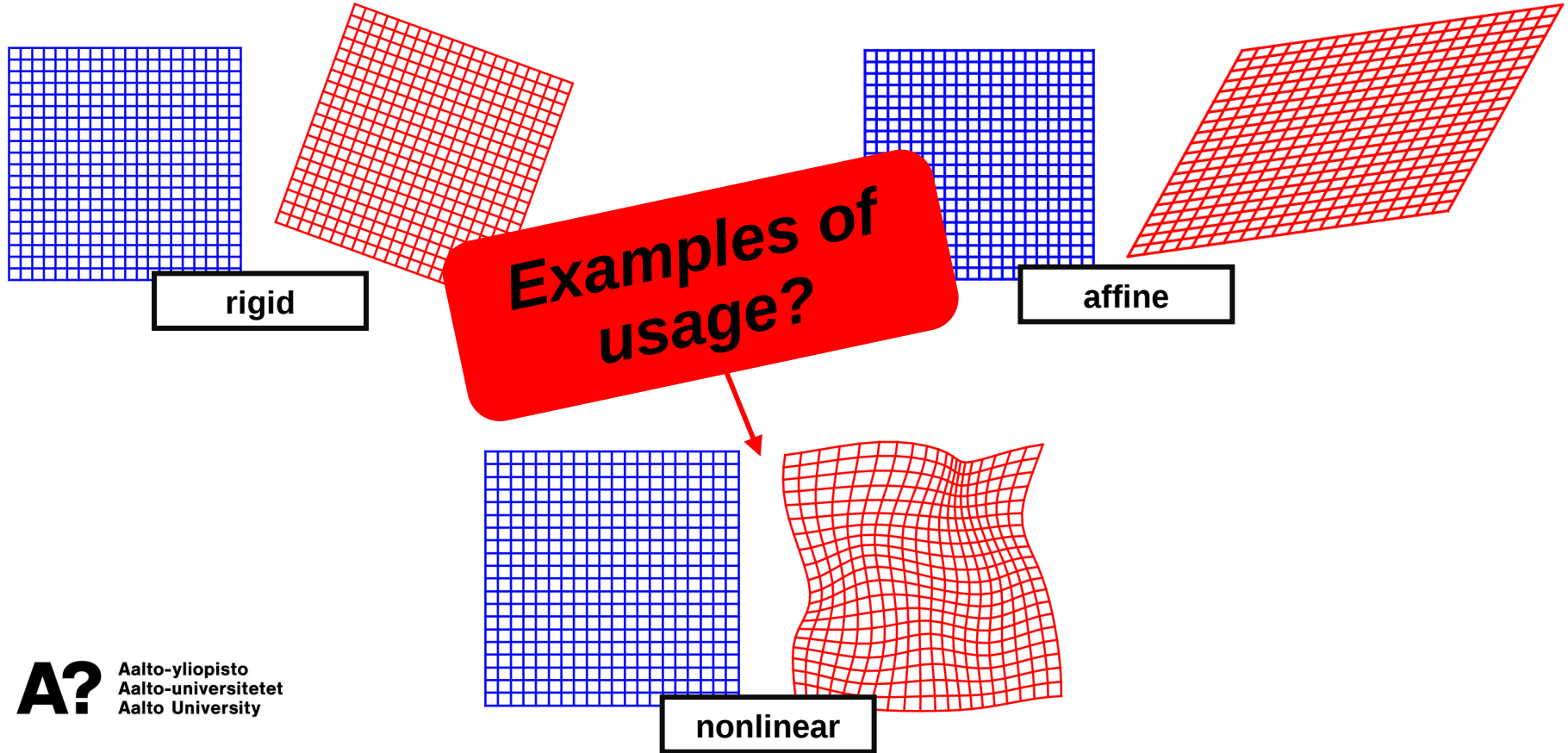
$$y_d(\mathbf{x}, \mathbf{w})$$

controls how points $\mathbf{x}$ in the fixed image
move along the d -th direction in the moving image
as the parameters $\mathbf{w}$ are varied

Spatial transformations



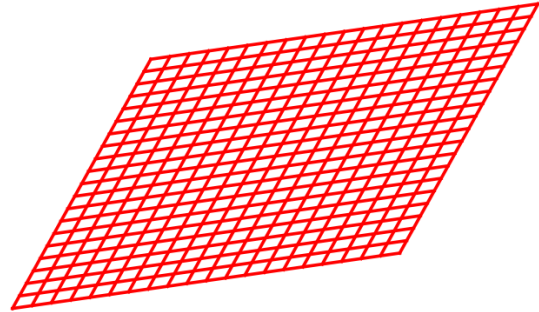
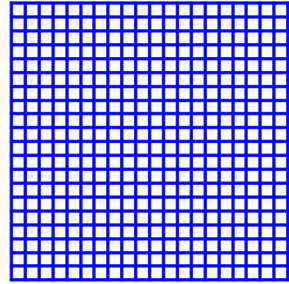
Spatial transformations



Affine transformation

$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{A}\mathbf{x} + \mathbf{t}$$

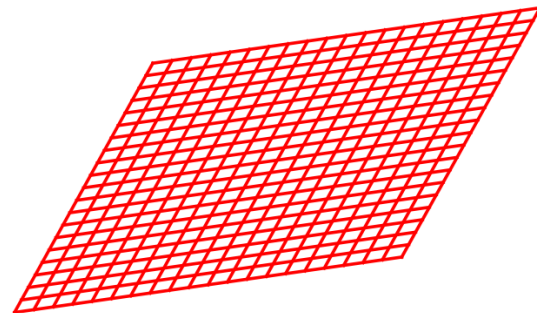
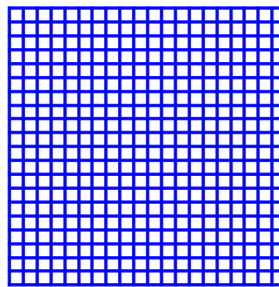
$$\mathbf{A} = \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} \quad \text{and} \quad \mathbf{t} = \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}$$



Affine transformation

$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{A}\mathbf{x} + \mathbf{t}$$

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$$y_d(\mathbf{x}, \mathbf{w})$$

controls how points $\mathbf{x}$ in the fixed image move along the d -th direction in the moving image as the parameters $\mathbf{w}$ are varied

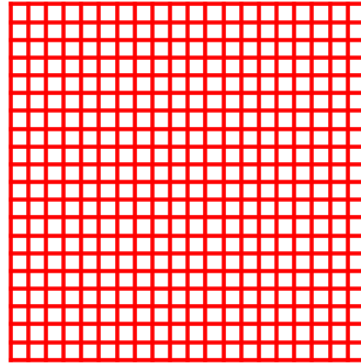
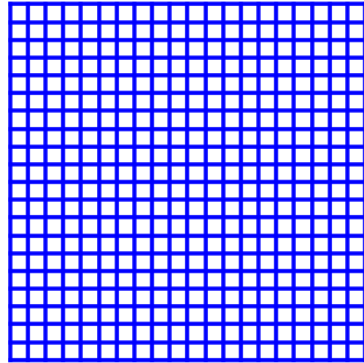
$$y_d(\mathbf{x}, \mathbf{w}_d) = t_d + a_{d,1}x_1 + \dots + a_{d,D}x_D$$

$$\mathbf{w}_d = (t_d, a_{d,1}, \dots, a_{d,D})^T$$

$$\mathbf{w} = (\mathbf{w}_1^T, \dots, \mathbf{w}_D^T)^T$$

Affine transformation

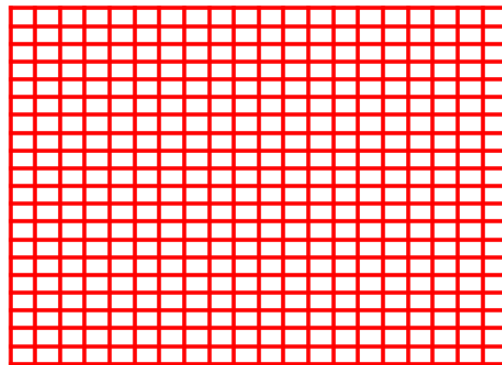
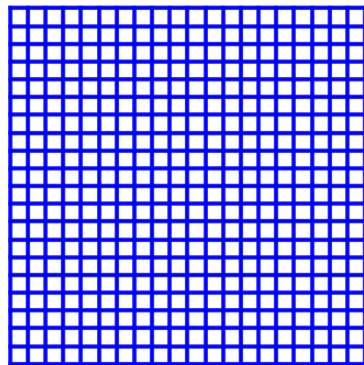
$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{A}\mathbf{x} + \mathbf{t}$$



$$\mathbf{A} = \begin{pmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{pmatrix}, \mathbf{t} = \begin{pmatrix} 23 \\ 0 \end{pmatrix}$$

Affine transformation

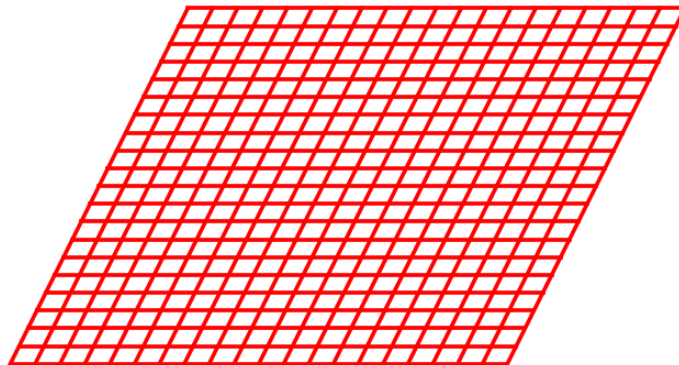
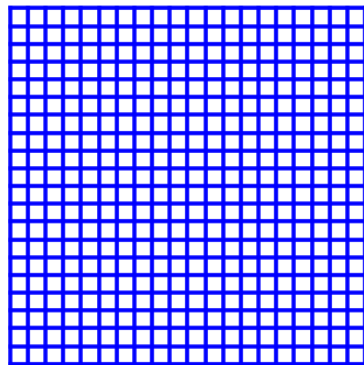
$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{A}\mathbf{x} + \mathbf{t}$$



$$\mathbf{A} = \begin{pmatrix} 1.4 & 0.0 \\ 0.0 & 1.0 \end{pmatrix}, \mathbf{t} = \begin{pmatrix} 23 \\ 0 \end{pmatrix}$$

Affine transformation

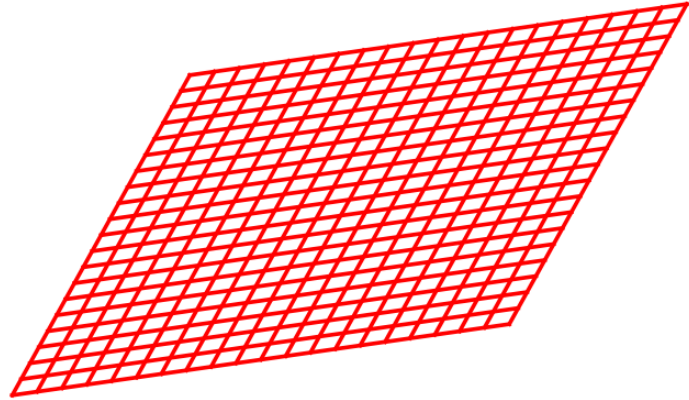
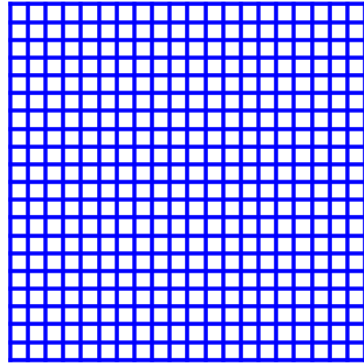
$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{A}\mathbf{x} + \mathbf{t}$$



$$\mathbf{A} = \begin{pmatrix} 1.4 & 0.5 \\ 0.0 & 1.0 \end{pmatrix}, \mathbf{t} = \begin{pmatrix} 23 \\ 0 \end{pmatrix}$$

Affine transformation

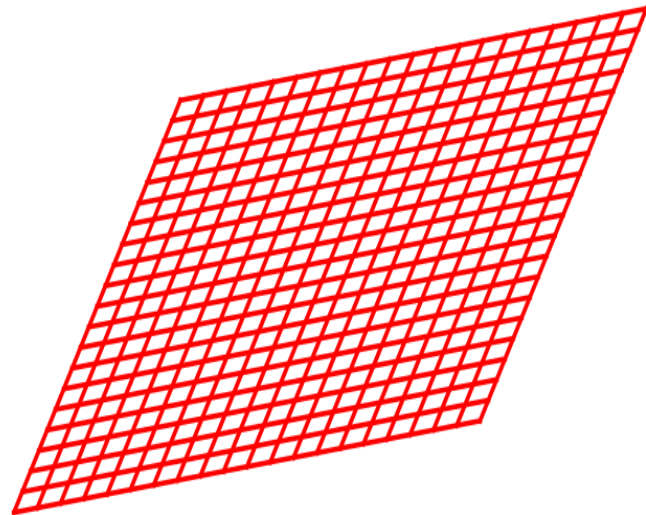
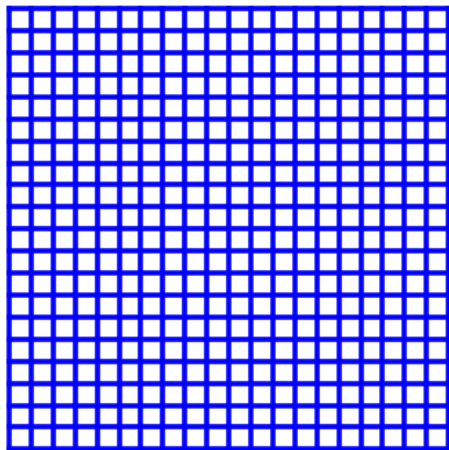
$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{A}\mathbf{x} + \mathbf{t}$$



$$\mathbf{A} = \begin{pmatrix} 1.4 & 0.5 \\ 0.2 & 0.9 \end{pmatrix}, \quad \mathbf{t} = \begin{pmatrix} 23 \\ 6 \end{pmatrix}$$

Affine (linear) transformation...

$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{A}\mathbf{x} + \mathbf{t}$$



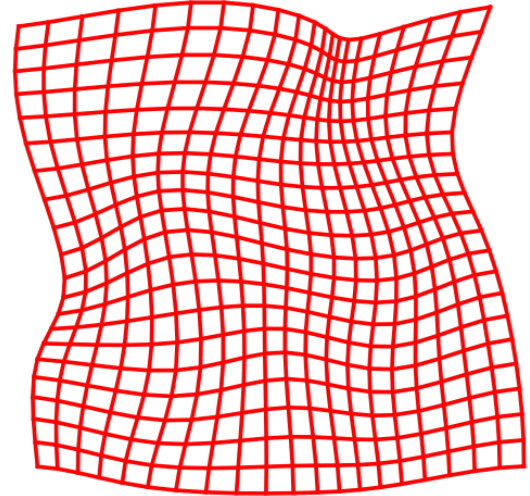
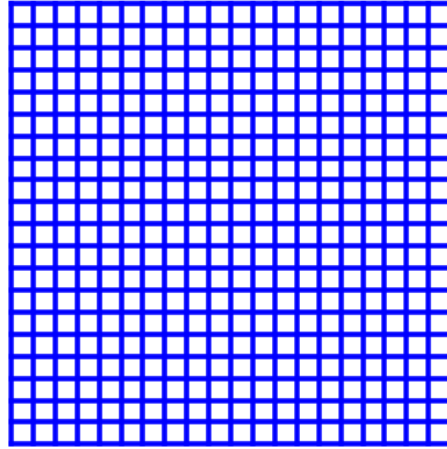
$$y_d(\mathbf{x}, \mathbf{w}_d) = t_d + a_{d,1}x_1 + \dots + a_{d,D}x_D$$

$$\mathbf{w}_d = (t_d, a_{d,1}, \dots, a_{d,D})^T$$

$$\mathbf{w} = (\mathbf{w}_1^T, \dots, \mathbf{w}_D^T)^T$$

...vs. nonlinear transformation

$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{x} + \delta(\mathbf{x}, \mathbf{w})$$



$$y_d(\mathbf{x}, \mathbf{w}_d) = x_d + \delta_d(\mathbf{x}, \mathbf{w}_d) \quad \text{with}$$

$$\mathbf{w}_d = (w_{d,0}, \dots, w_{d,M-1})^T$$

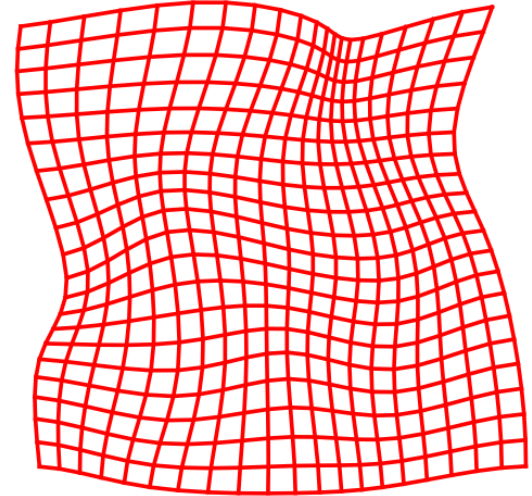
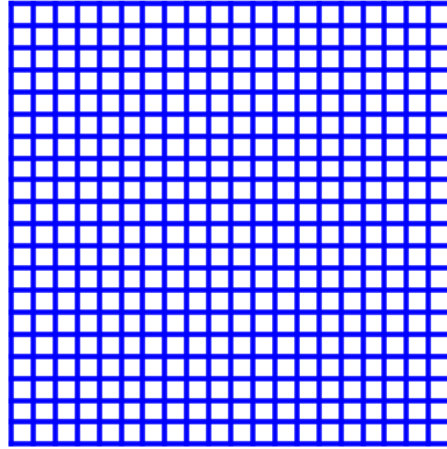
$$\mathbf{w} = (\mathbf{w}_1^T, \dots, \mathbf{w}_D^T)^T$$

$$\delta_d(\mathbf{x}, \mathbf{w}_d) = \sum_{m=0}^{M-1} w_{d,m} \phi_m(\mathbf{x})$$

...vs. nonlinear transformation

“residual deformation”

$$\mathbf{y}(\mathbf{x}, \mathbf{w}) = \mathbf{x} + \delta(\mathbf{x}, \mathbf{w})$$



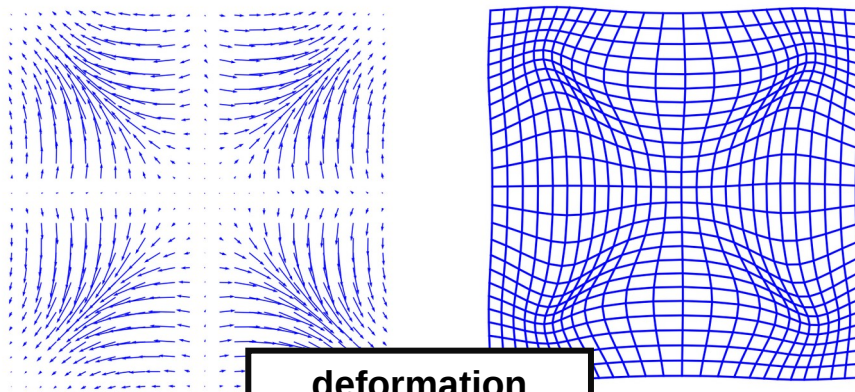
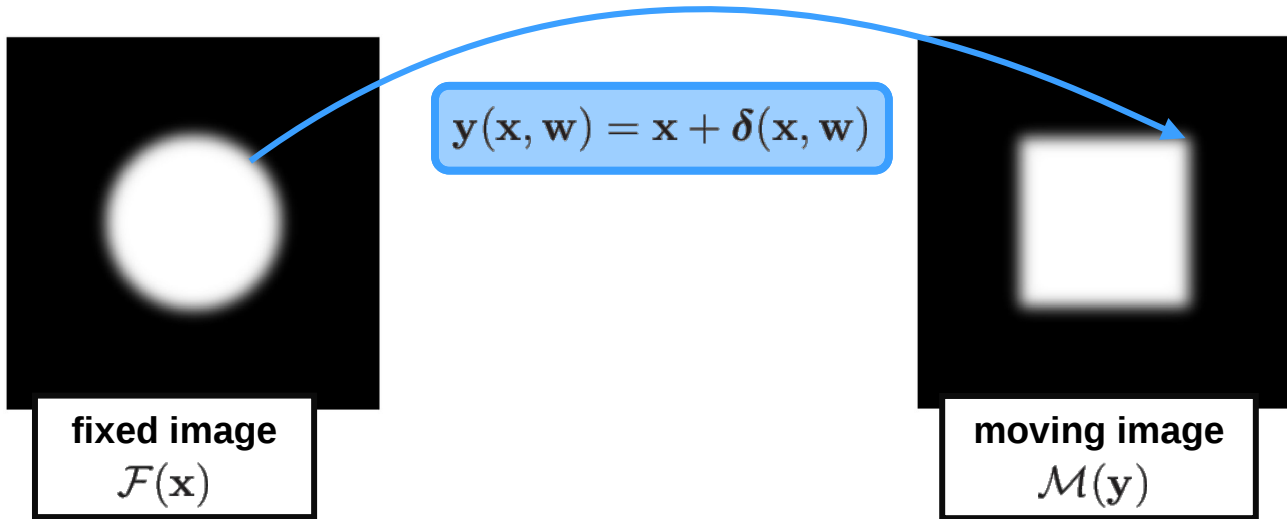
$$\mathbf{y}_d(\mathbf{x}, \mathbf{w}_d) = \mathbf{x}_d + \delta_d(\mathbf{x}, \mathbf{w}_d) \quad \text{with}$$

$$\mathbf{w}_d = (w_{d,0}, \dots, w_{d,M-1})^T$$

$$\mathbf{w} = (\mathbf{w}_1^T, \dots, \mathbf{w}_D^T)^T$$

$$\delta_d(\mathbf{x}, \mathbf{w}_d) = \sum_{m=0}^{M-1} w_{d,m} \phi_m(\mathbf{x})$$

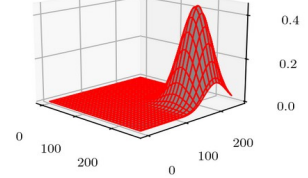
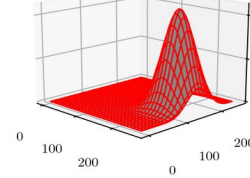
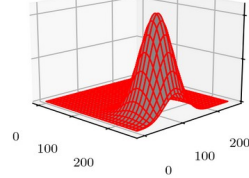
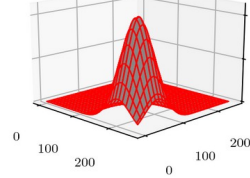
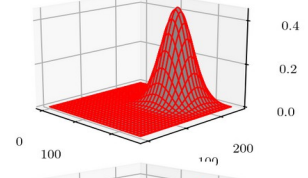
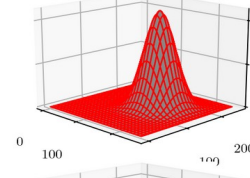
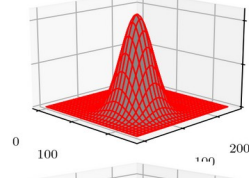
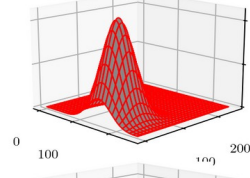
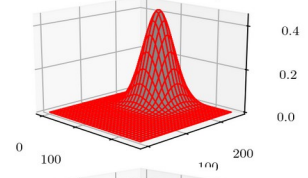
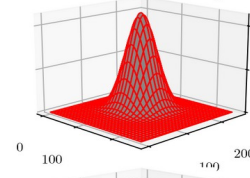
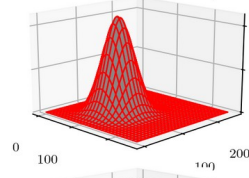
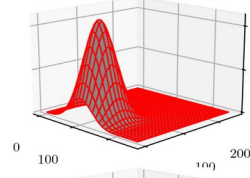
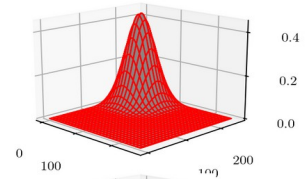
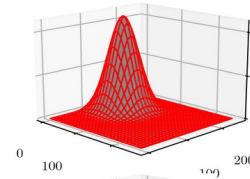
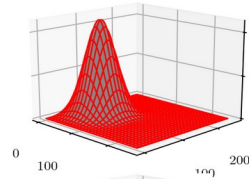
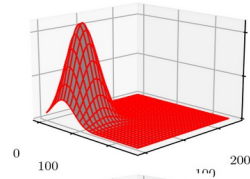
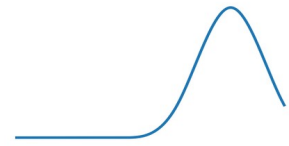
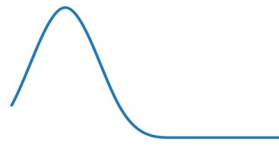
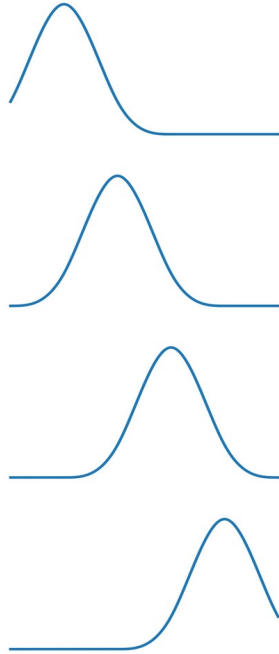
nonlinear basis
functions

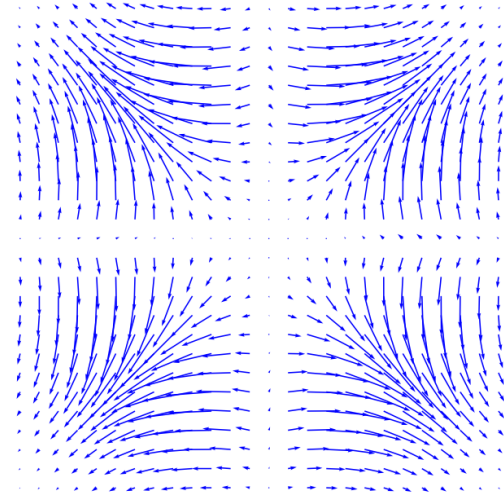
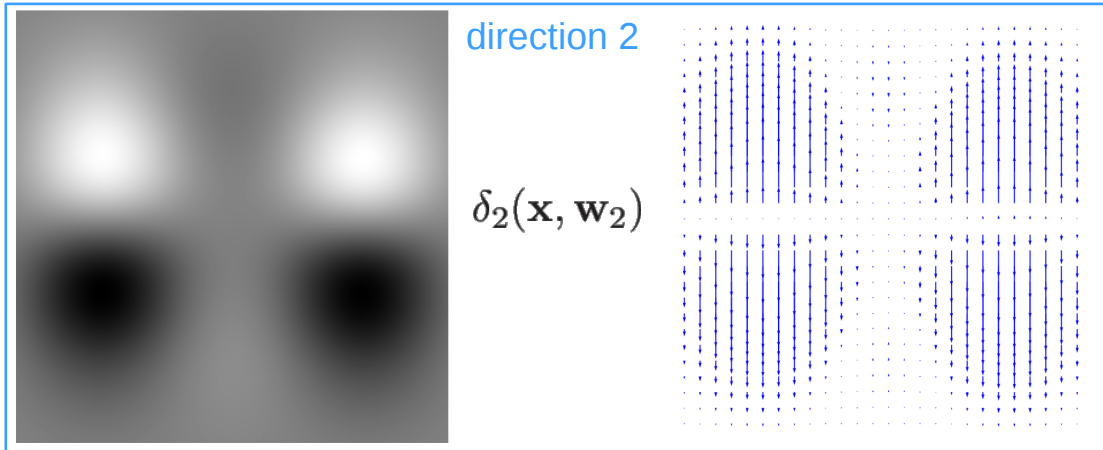
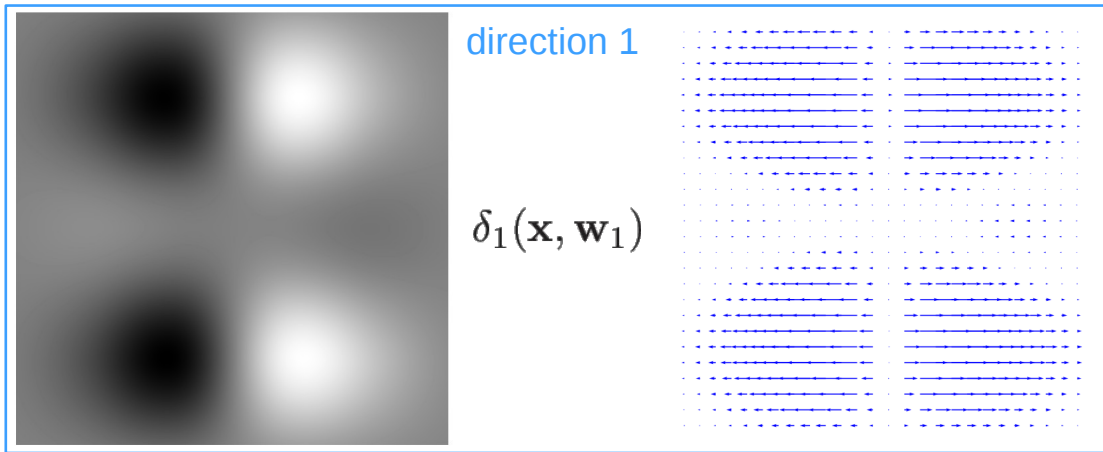


deformation
 $\delta(\mathbf{x}, \mathbf{w})$



interpolated moving image
 $\mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))$

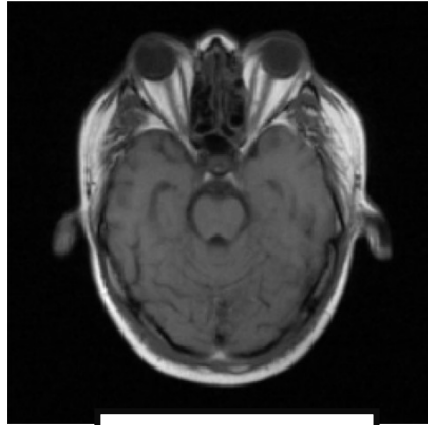




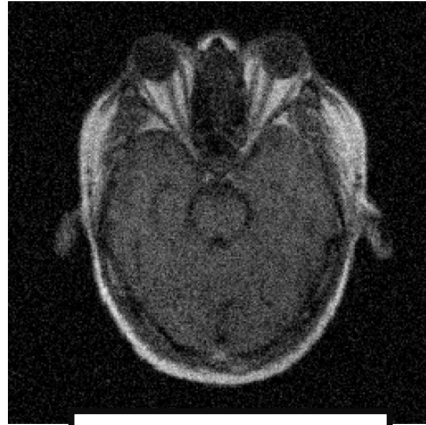
deformation
 $\delta(\mathbf{x}, \mathbf{w})$

Focus on intra-modal registration

Images have similar intensity characteristics



$\mathcal{F}(\mathbf{x})$



$\mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))$

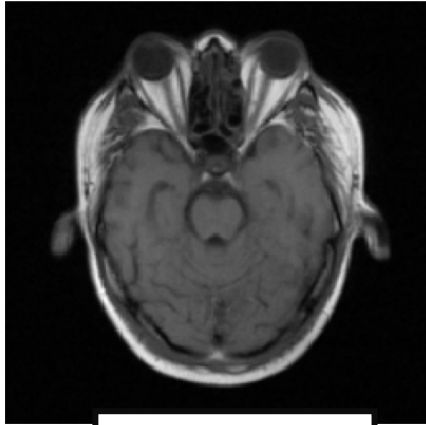


$[\mathcal{F}(\mathbf{x}) - \mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))]^2$

$$E(\mathbf{w}) = \sum_{n=1}^N [\mathcal{F}(\mathbf{x}_n) - \mathcal{M}(\mathbf{y}(\mathbf{x}_n, \mathbf{w}))]^2$$

Focus on intra-modal registration

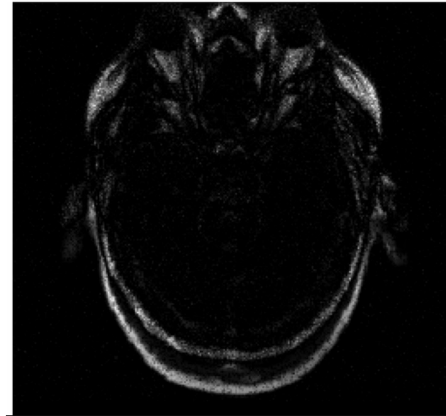
Images have similar intensity characteristics



$\mathcal{F}(\mathbf{x})$

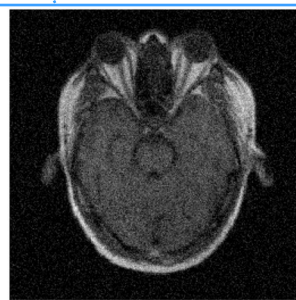
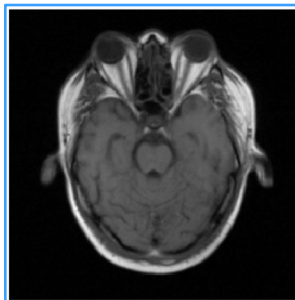
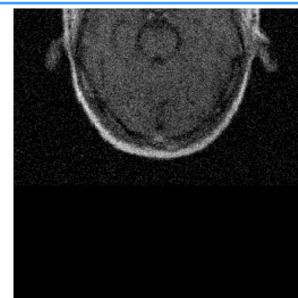
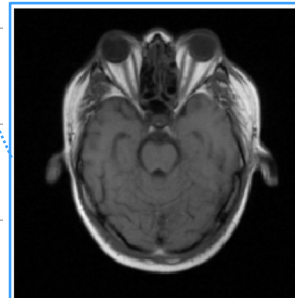
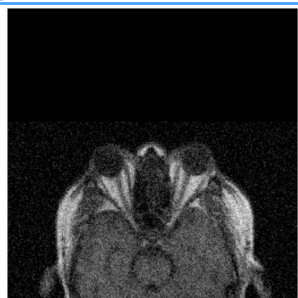
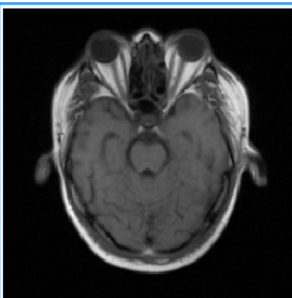
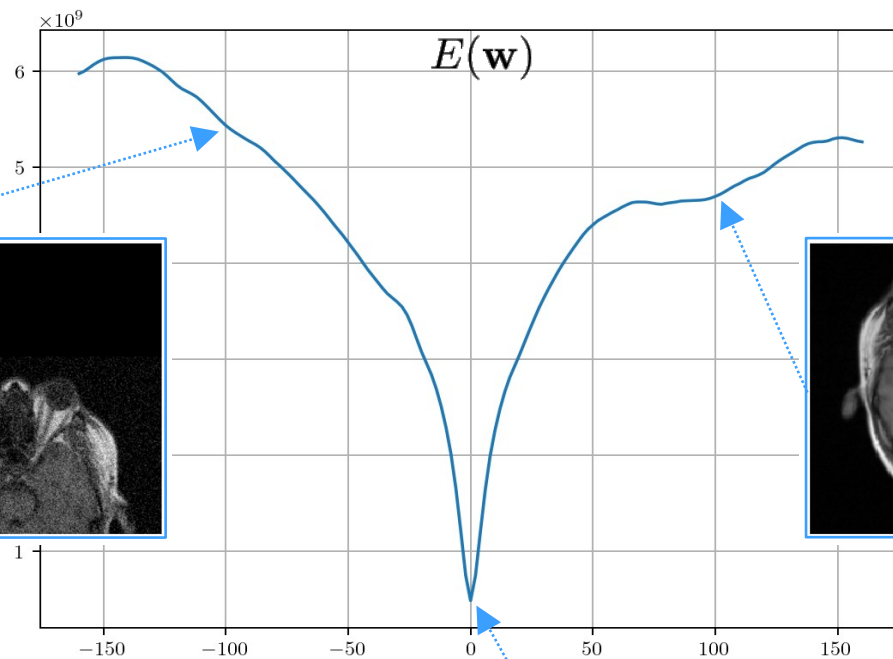


$\mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))$



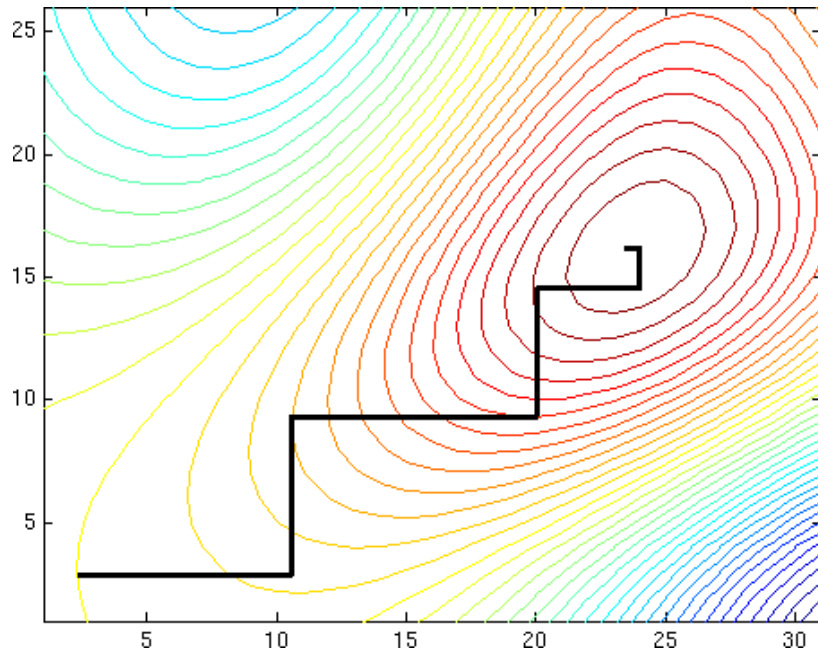
$[\mathcal{F}(\mathbf{x}) - \mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))]^2$

$$E(\mathbf{w}) = \sum_{n=1}^N [\mathcal{F}(\mathbf{x}_n) - \mathcal{M}(\mathbf{y}(\mathbf{x}_n, \mathbf{w}))]^2$$



Numerical optimization

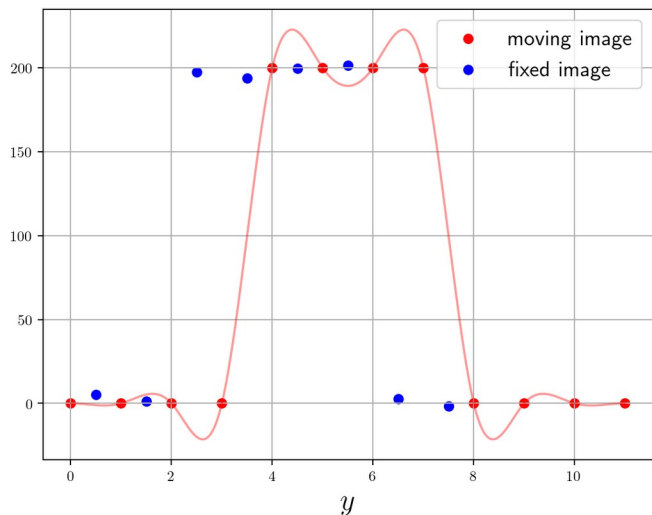
Find transformation parameters $\mathbf{w}$ that minimize $E(\mathbf{w})$



Gauss-Newton optimization

Toy example: 1D and translation only

- ✓ transformation model: $y(x, t) = x + t$ parameter to be optimized
- ✓ energy: $E(t) = \sum_{n=1}^N E_n(t)$ with $E_n(t) = [\mathcal{F}(x) - \mathcal{M}(y(x_n, t))]^2$

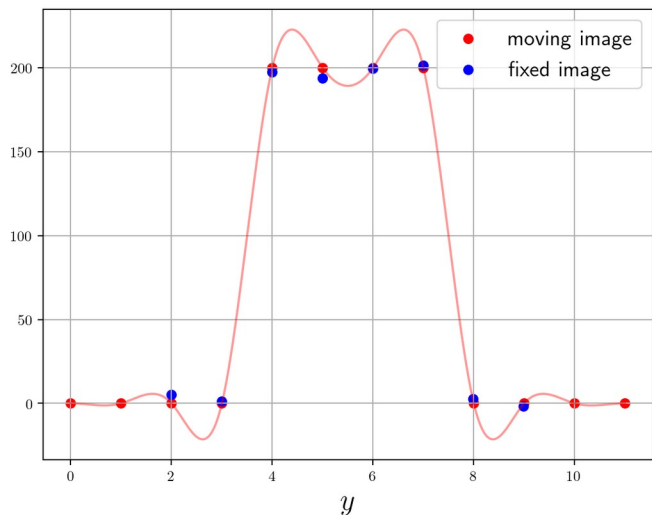


before optimization

Gauss-Newton optimization

Toy example: 1D and translation only

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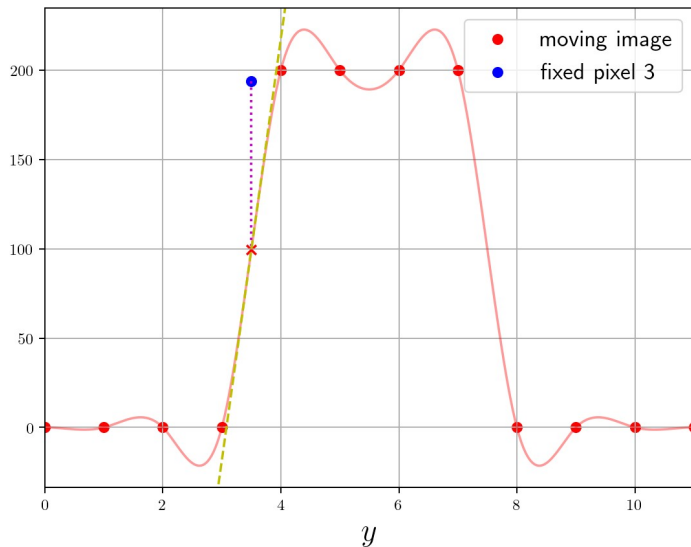


after optimization

Gauss-Newton optimization

$$g_n = \left. \frac{d\mathcal{M}(y)}{dy} \right|_{y=x_n+t}$$

Idea: for a small deviation ϵ around current estimate of t : $\mathcal{M}(y(x_n, t + \epsilon)) \simeq \mathcal{M}(y(x_n, t)) + g_n \cdot \epsilon$



✓ Energy: $E(t + \epsilon) = \sum_{n=1}^N E_n(t + \epsilon)$ with $E_n(t + \epsilon) = [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t + \epsilon))]^2$
 $\simeq [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t)) - g_n \cdot \epsilon]^2$

Task: what is ϵ minimizing $E(t + \epsilon)$?

Gauss-Newton optimization

Solution: standard linear regression!

$$\epsilon = (\psi^T \psi)^{-1} \psi^T \tau$$

Now update t :

$$t \leftarrow t + \epsilon$$

where $\tau = \begin{pmatrix} \tau_1 \\ \vdots \\ \tau_N \end{pmatrix}$ with $\tau_n = \mathcal{F}(x_n) - \mathcal{M}(y(x_n, t))$

and $\psi = \begin{pmatrix} g_1 \\ \vdots \\ g_N \end{pmatrix}$

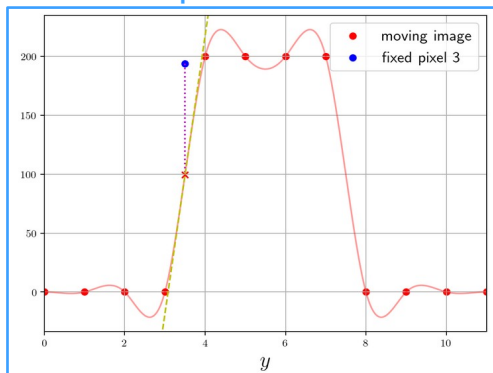
one basis function

initialization

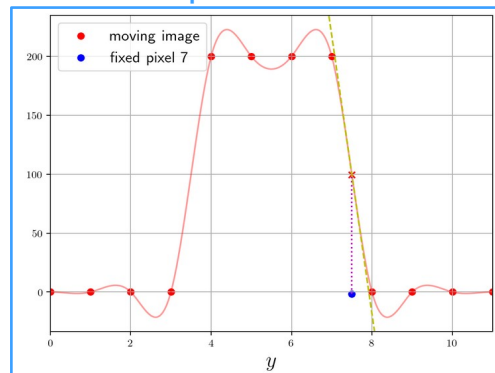
pixel 3

pixel 7

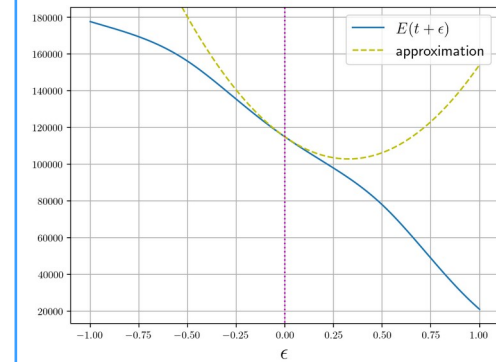
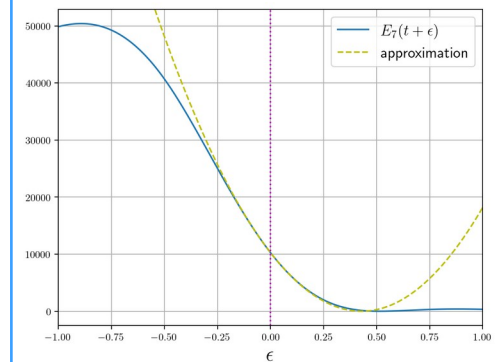
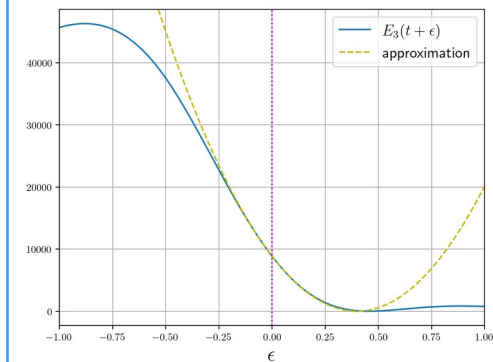
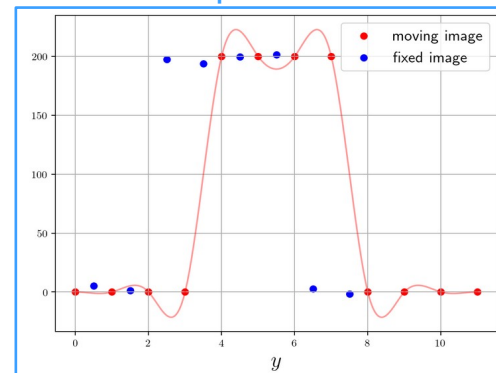
all pixels



... + ...



+ ... =



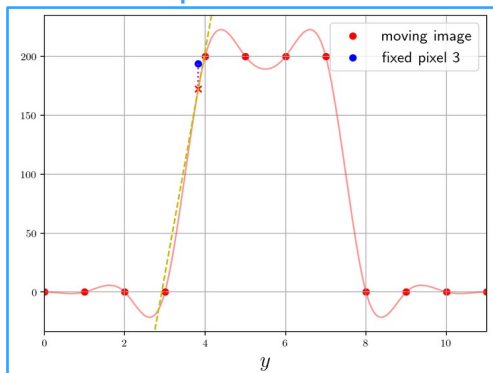
✓ Energy: $E(t + \epsilon) = \sum_{n=1}^N E_n(t + \epsilon)$ with $E_n(t + \epsilon) = [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t + \epsilon))]^2$
 $\simeq [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t)) - g_n \cdot \epsilon]^2$

after 1 iteration

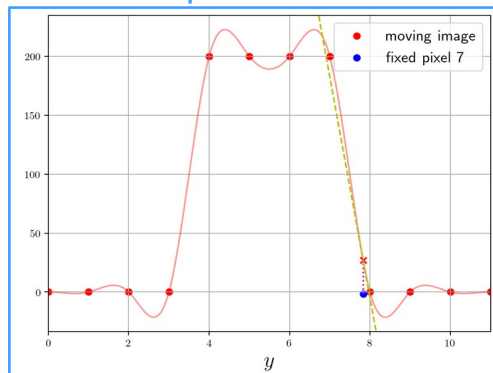
pixel 3

pixel 7

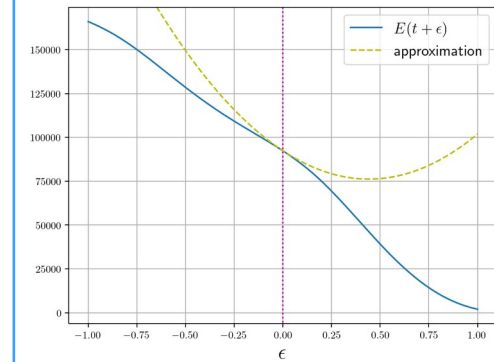
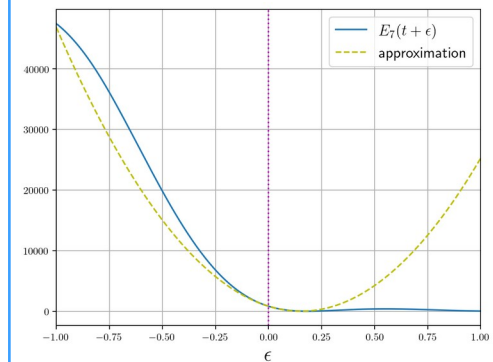
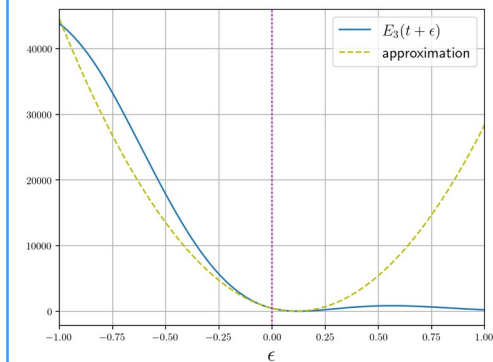
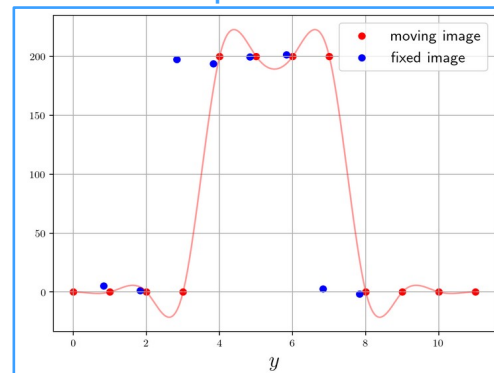
all pixels



... + ...



+ ... =



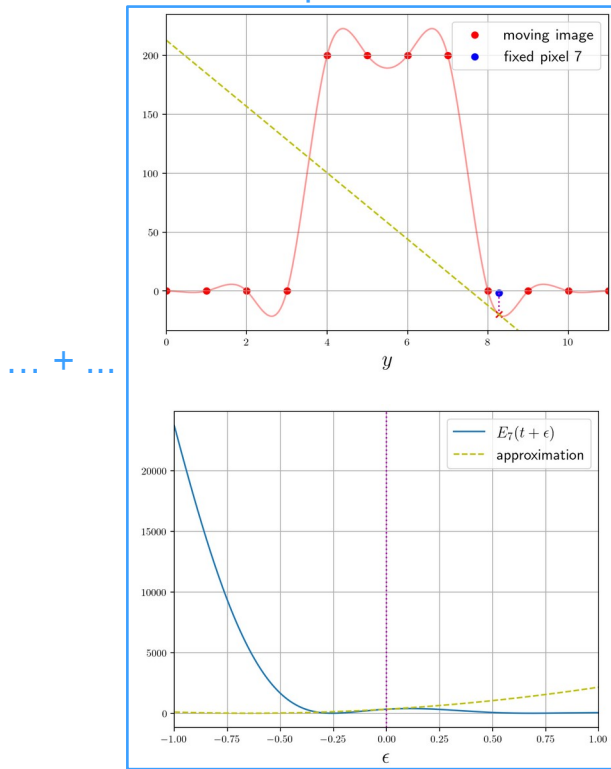
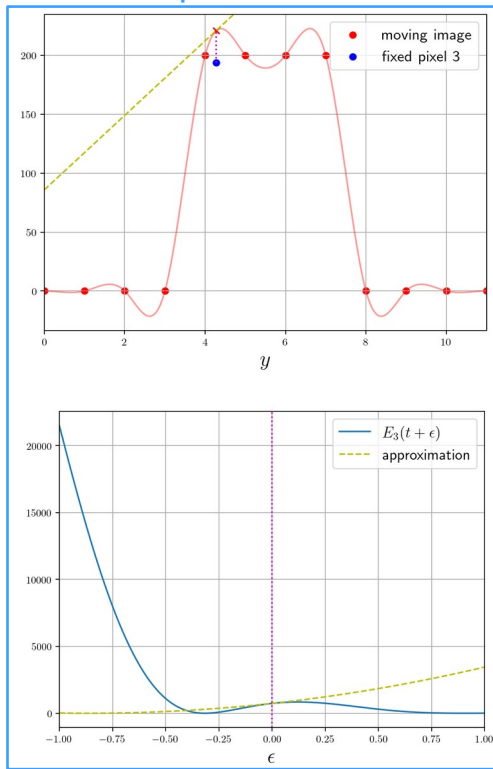
✓ Energy: $E(t + \epsilon) = \sum_{n=1}^N E_n(t + \epsilon)$ with $E_n(t + \epsilon) = [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t + \epsilon))]^2$
 $\simeq [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t)) - g_n \cdot \epsilon]^2$

after 2 iterations

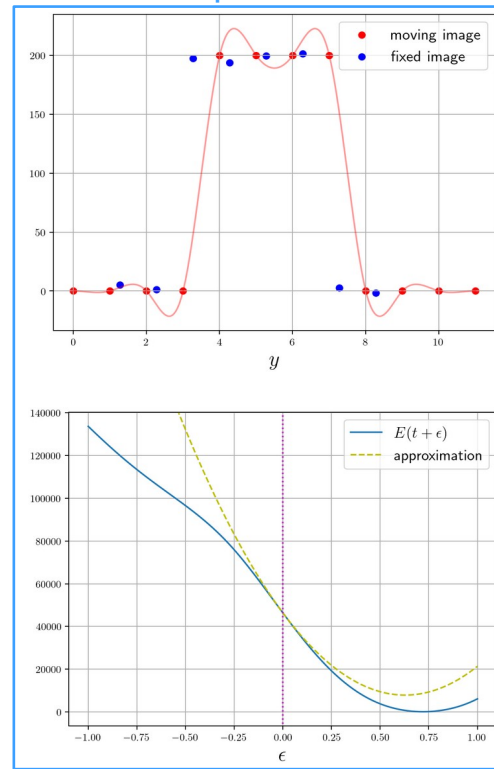
pixel 3

pixel 7

all pixels



+ ... =



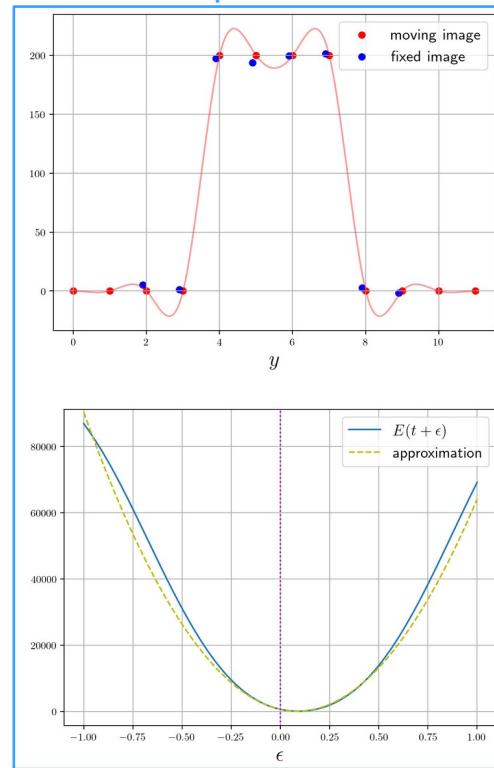
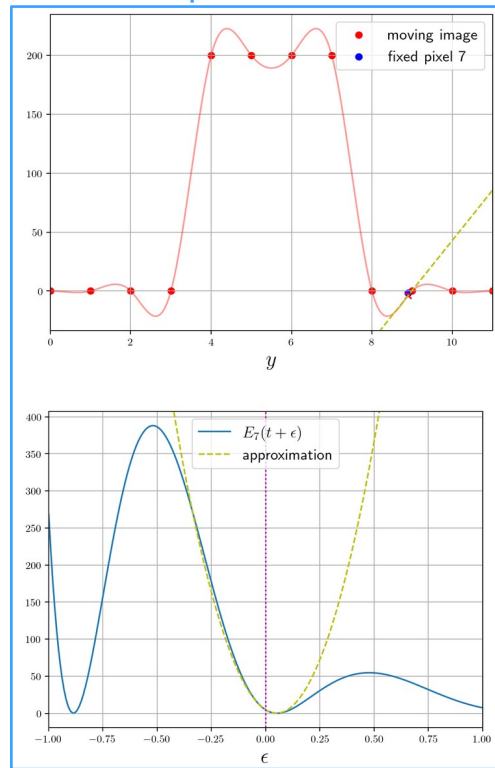
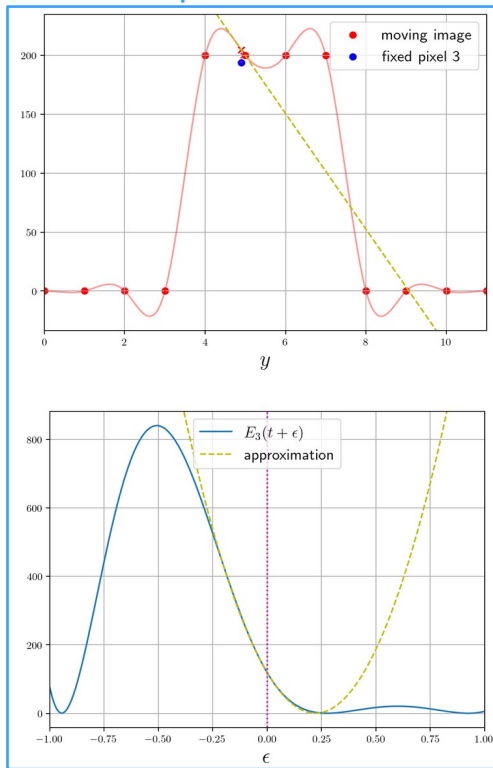
✓ Energy: $E(t + \epsilon) = \sum_{n=1}^N E_n(t + \epsilon)$ with $E_n(t + \epsilon) = [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t + \epsilon))]^2$
 $\simeq [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t)) - g_n \cdot \epsilon]^2$

after 3 iterations

pixel 3

pixel 7

all pixels



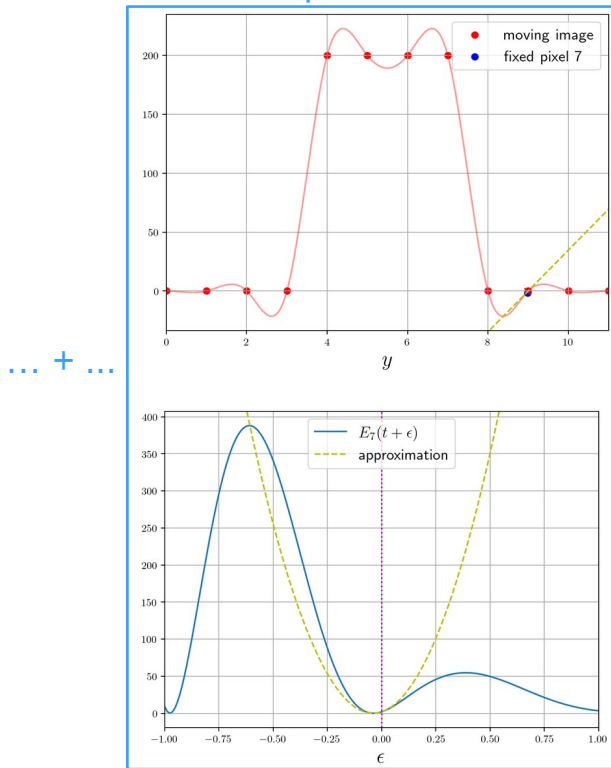
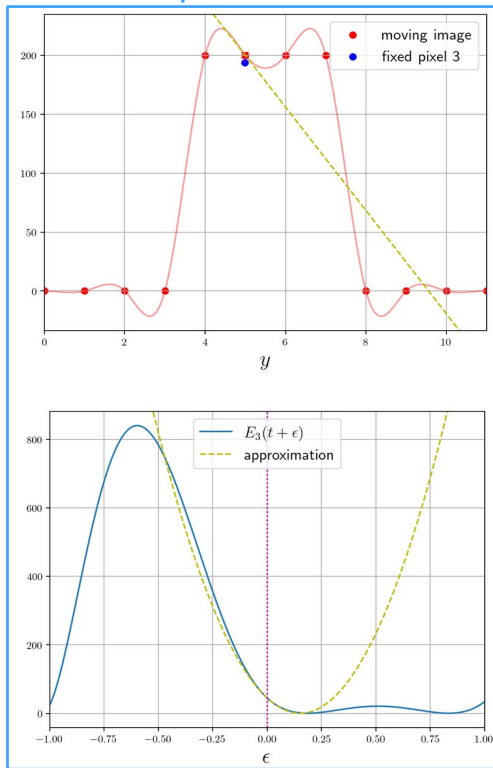
✓ Energy: $E(t + \epsilon) = \sum_{n=1}^N E_n(t + \epsilon)$ with $E_n(t + \epsilon) = [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t + \epsilon))]^2$
 $\simeq [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t)) - g_n \cdot \epsilon]^2$

after 4 iterations

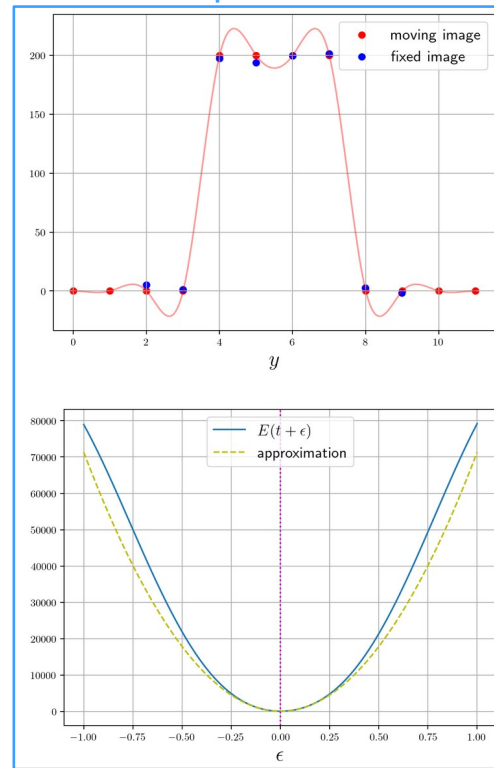
pixel 3

pixel 7

all pixels



+ ... =



✓ Energy: $E(t + \epsilon) = \sum_{n=1}^N E_n(t + \epsilon)$ with $E_n(t + \epsilon) = [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t + \epsilon))]^2$
 $\simeq [\mathcal{F}(x_n) - \mathcal{M}(y(x_n, t)) - g_n \cdot \epsilon]^2$

Gauss-Newton optimization

Solution: standard linear regression!

$$\epsilon = (\psi^T \psi)^{-1} \psi^T \tau$$

Now update t :
 $t \leftarrow t + \epsilon$

where $\tau = \begin{pmatrix} \tau_1 \\ \vdots \\ \tau_N \end{pmatrix}$ with $\tau_n = \mathcal{F}(x_n) - \mathcal{M}(y(x_n, t))$

and $\psi = \begin{pmatrix} g_1 \\ \vdots \\ g_N \end{pmatrix}$

one basis function

Gauss-Newton optimization

Solution: standard linear regression!

Now update t :
 $t \leftarrow t + \epsilon$

$$\epsilon = (\psi^T \psi)^{-1} \psi^T \tau$$

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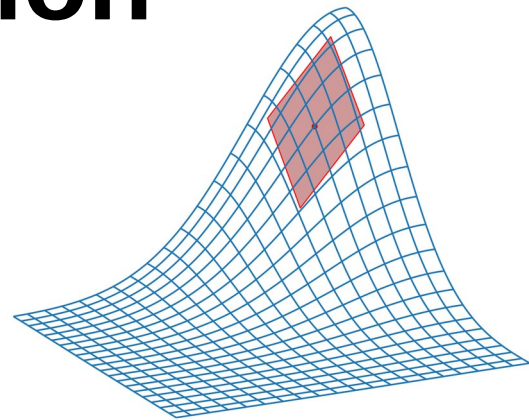
and $\psi = \begin{pmatrix} g_1 \\ \vdots \\ g_N \end{pmatrix} = \underbrace{\begin{pmatrix} g_1 & & \\ & \ddots & \\ & & g_N \end{pmatrix}}_G \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$

one basis function

Gauss-Newton optimization

In real situations, exactly the same idea but:

- ✓ $D > 1$ spatial dimensions
- ✓ $M > 1$ basis functions



$$\mathcal{M}(\mathbf{y}(\mathbf{x}_n, \mathbf{w} + \boldsymbol{\epsilon})) \simeq \mathcal{M}(\mathbf{y}(\mathbf{x}_n, \mathbf{w})) + \sum_{d=1}^D \sum_{m=0}^M (g_{d,n} \phi_m(\mathbf{x}_n)) \epsilon_{d,m}$$

Solution: $\boldsymbol{\epsilon} = \left(\boldsymbol{\Psi}^T \boldsymbol{\Psi} \right)^{-1} \boldsymbol{\Psi}^T \boldsymbol{\tau}$ where $\boldsymbol{\Psi} = \left(\mathbf{G}_1 \boldsymbol{\Phi} \mid \cdots \mid \mathbf{G}_D \boldsymbol{\Phi} \right)$

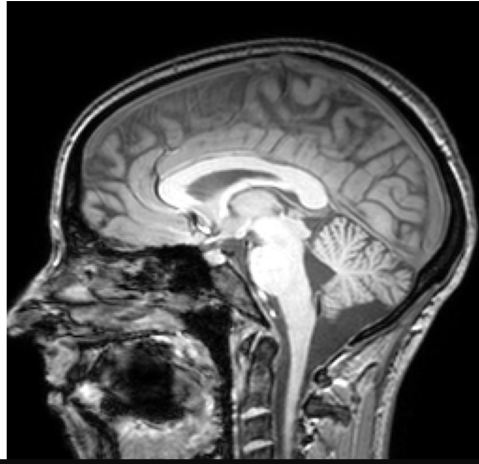
and $\boldsymbol{\Phi} = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & \cdots & \phi_{M-1}(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{pmatrix}$

initialization



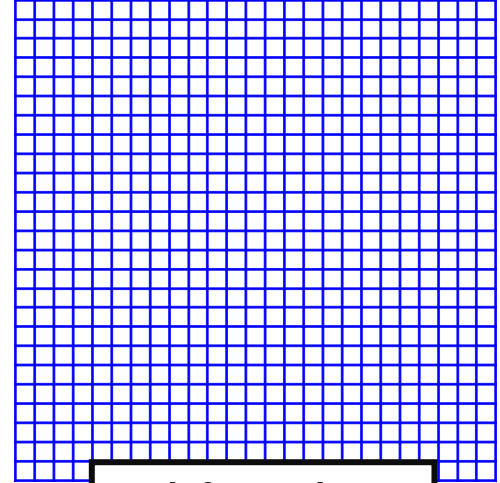
fixed image

$$\mathcal{F}(\mathbf{x})$$



interpolated moving image

$$\mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))$$



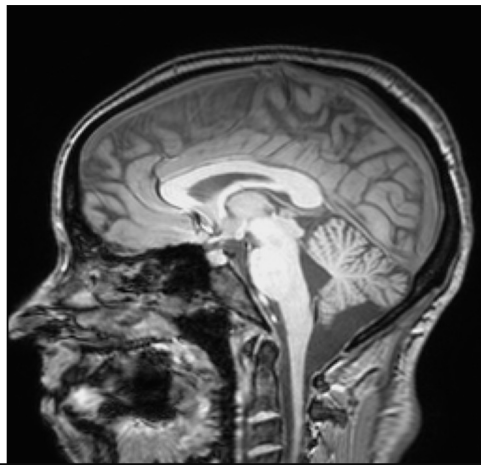
deformation

$$\delta(\mathbf{x}, \mathbf{w})$$

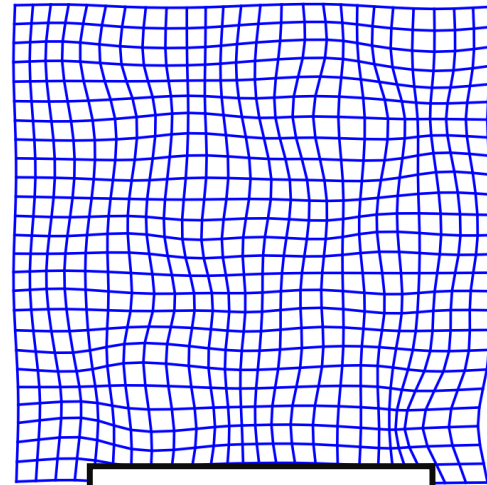
after 10 iterations



fixed image
 $\mathcal{F}(\mathbf{x})$



interpolated moving image
 $\mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))$

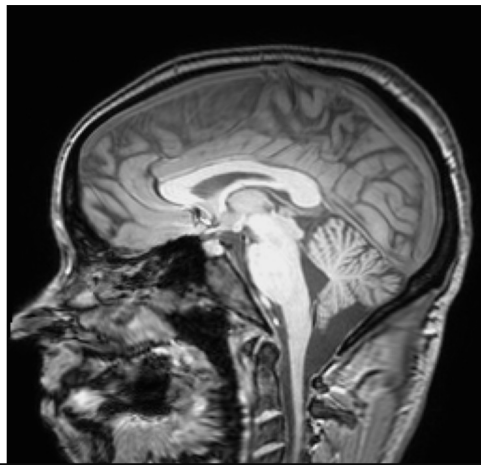


deformation
 $\delta(\mathbf{x}, \mathbf{w})$

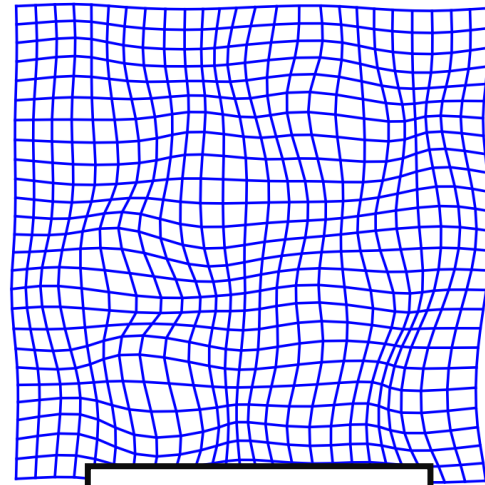
after 30 iterations



fixed image
 $\mathcal{F}(\mathbf{x})$



interpolated moving image
 $\mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))$



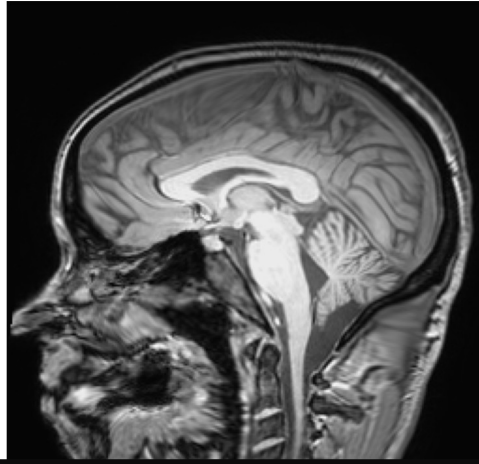
deformation
 $\delta(\mathbf{x}, \mathbf{w})$

after convergence



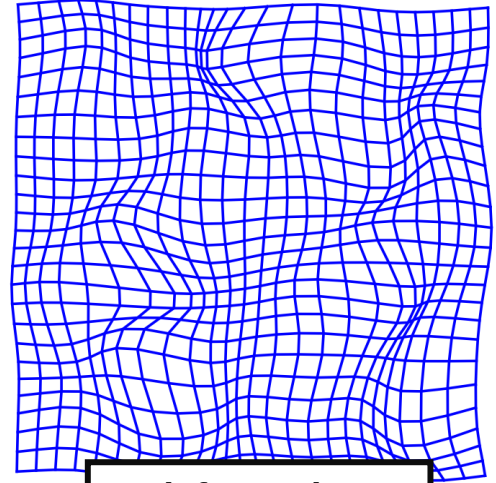
fixed image

$$\mathcal{F}(\mathbf{x})$$



interpolated moving image

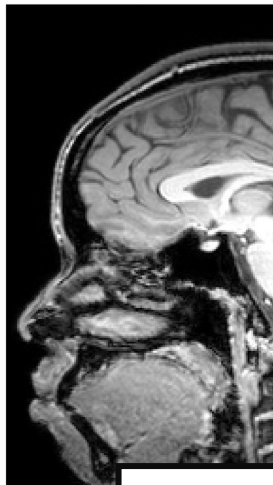
$$\mathcal{M}(\mathbf{y}(\mathbf{x}, \mathbf{w}))$$



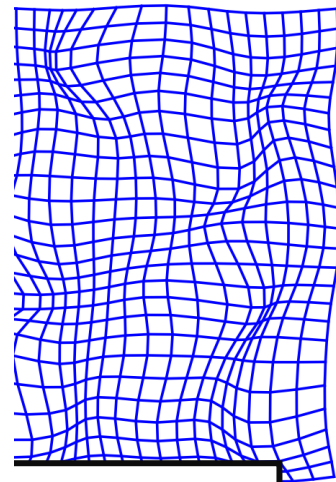
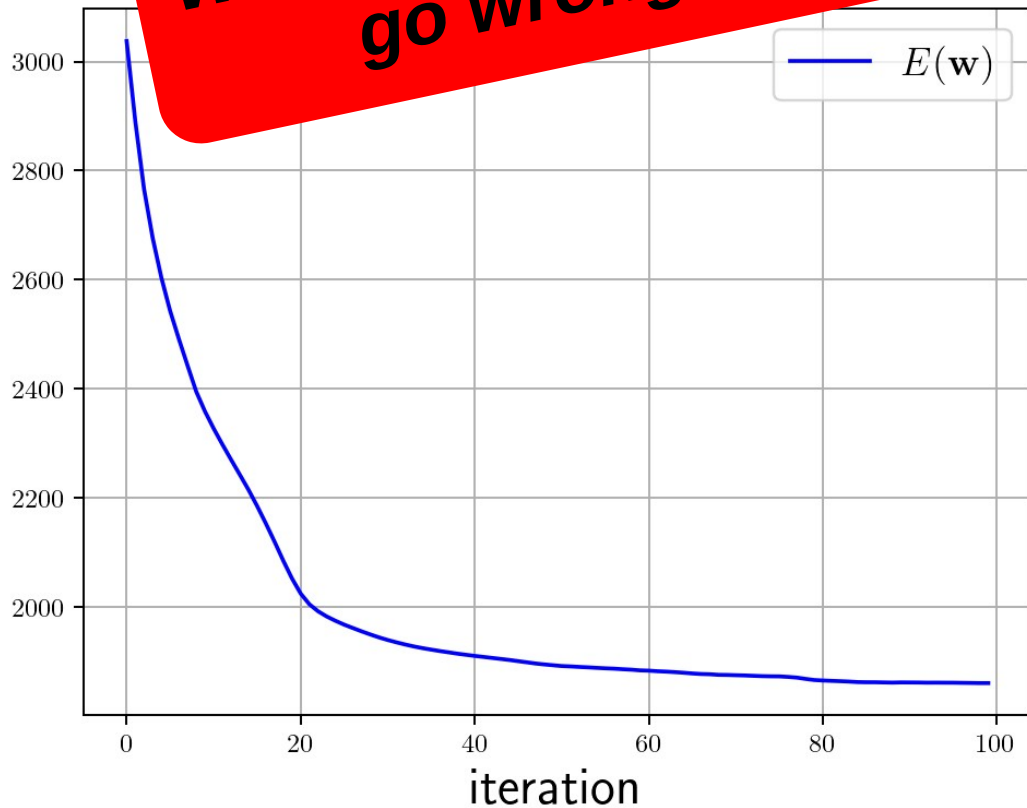
deformation

$$\delta(\mathbf{x}, \mathbf{w})$$

What could possibly go wrong?



fixed input
 $\mathcal{F}(\mathbf{x})$



Information
 $\delta(\mathbf{x}, \mathbf{w})$

Safety rails...

We have assumed that ϵ is small

- ✓ What if it isn't? Energy $E(\mathbf{w})$ could go up instead of down!
- ✓ Solution: Levenberg–Marquardt

$$\epsilon = \left(\Psi^T \Psi + \lambda \mathbf{I} \right)^{-1} \Psi^T \tau$$

tunable

